

Three systems, three answers

One Pacific chart carries a trough that may become something, a low that has not formed, and a depression that already has. The archive is asked the same question of each — **what did storms born here, in September, go on to do?** — and gives three different kinds of answer. **The most useful of them is a refusal.**

D1

Trough S-SE of Hawai'i

OUTLOOK AREA · BROAD TROUGH, NO CENTRE

RATES REFUSED

3 analogs · ESS 2.5 · gate 10

D1 — the archive declines

13.2N 147.3W

NHC gives this trough **10% / 20%**. The pool is **3 storms** — effective sample **2.5** against a gate of 10. Below that line the archive publishes counts and refuses rates.

TD	3/3
TS	1/3
Cat 1	0/3
Cat 2	0/3
Cat 3	0/3
Cat 4	0/3

Counts out of the pool. No rate is drawn because none is published at this sample size.

ANALYST-DECLARED REFERENCE CELL FROM NHC FORECAST AREA

Not an observed centre, not an Invest, not a disturbance position, not a genesis point.

D2

Low forecast to form

FORECAST FORMATION AREA · NO CURRENT DISTURBANCE

CONDITIONAL RATES PUBLISHED · NO COMPOSED NUMBER

15 analogs · ESS 10.5 · gate 10

D2 — no unconditional number

15.8N 120.0W

TD	100%
TS	87%
Cat 1	47%
Cat 2	27%
Cat 3	13%
Cat 4	13%

Reached each threshold, 15 storms, 95% Wilson intervals over distinct storms. Conditional on formation under the declared reference condition.

NO COMPOSED NUMBER

NHC publishes a formation probability for its forecast area. Storm Atlas publishes historical outcome frequencies conditional on formation under the declared reference condition. Because those conditioning events are not identical, the two are not multiplied here.

ANALYST-DECLARED REFERENCE CELL FROM NHC FORECAST AREA

Not an observed centre, not an Invest, not a disturbance position, not a genesis point.

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TD Fourteen-E

OBSERVED GENESIS POINT · ADVISORIES RUNNING

CONDITIONAL RATES + HISTORICAL TIMING

23 analogs · ESS 18.2 · gate 10

Fourteen-E — genesis 15.8N 115.0W

rates and timing

TD	100%
TS	91%
Cat 1	30%
Cat 2	17%
Cat 3	9%
Cat 4	4%

Given genesis here · 23 storms, effective sample 18.2. It has formed, so no formation chance is needed.

GENESIS ·	FORECAST	ELAPSED
FIRST TROPICAL	65 KT	72 h
PT	12/1800Z	
09/1800Z		

forecast +72h

med +42h

+0h

+48h

+96h

+144h

Of 7 analogs reaching Cat 1: median +42 h, p90 +150 h. The forecast's +72 h is above p10, p25, median and below p75, p90.

Why D1's pool is smaller still

all-eras gap

3 of 6 analogs are from before 1971, when East Pacific intensities were estimated without geostationary satellites or Dvorak analysis and major hurricanes were under-observed (measured: 1.7% Cat 3 in the 1960s vs 20-30% from the 1970s on).

Restricting to 1971+ takes the pool from 6 to 3. Both are below the gate, so the refusal stands either way.

What this specimen corrects

vs. first draft

No composed probability. An earlier draft multiplied NHC's 50% by the archive's conditional rate. Withdrawn: the conditioning events are not the same event.

Clocks aligned. Timing was measured from the DB first-track fix (120 h) against a genesis-measured cohort, and called beyond p90. Withdrawn — on the shared clock it is 72 h, inside the distribution.

Points typed. An outlook centroid is an analyst-declared reference cell, not an observed genesis point.

Nearest analogs

by distance from genesis

SEASON	NAME	KM	PEAK	LANDFALL
2018	PAUL	63	40 kt	—
2009	NORA	106	50 kt	—
2001	KIKO	120	65 kt	—
2005	JOVA	123	110 kt	—

Weight falls with distance from genesis; it drives the effective sample size and is not a probability.